

Expert Systems

Capturing Human Expertise in AI

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- Introduction to AI and Expert Systems
 - Architecture of Expert Systems
 - Roles of individuals who interact with expert systems
 - Development Languages and Tools
 - Notable Examples and Applications
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Artificial intelligence

**Creating machines that simulate
human intelligence**

Machine Learning

computer vision

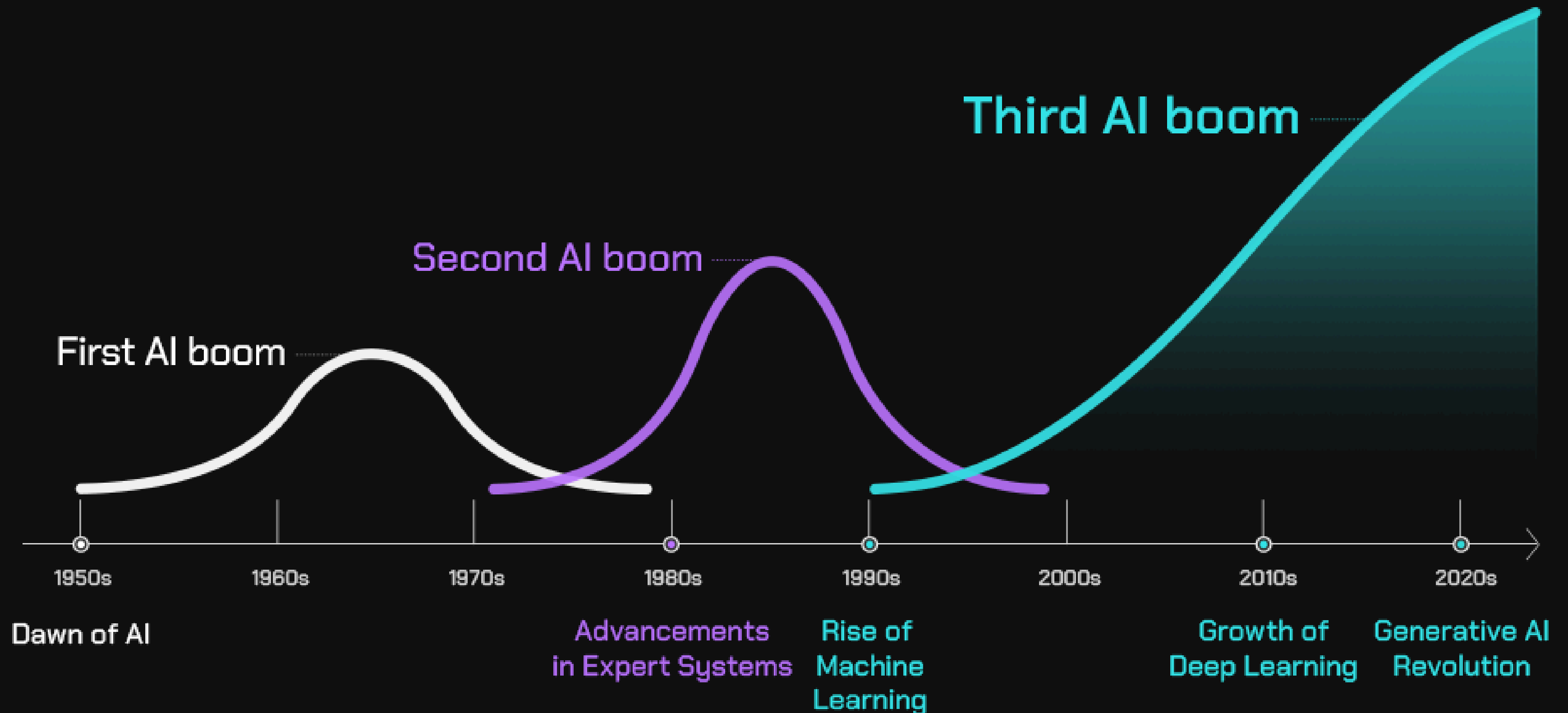
natural language processing

robotics

Transformers

expert systems

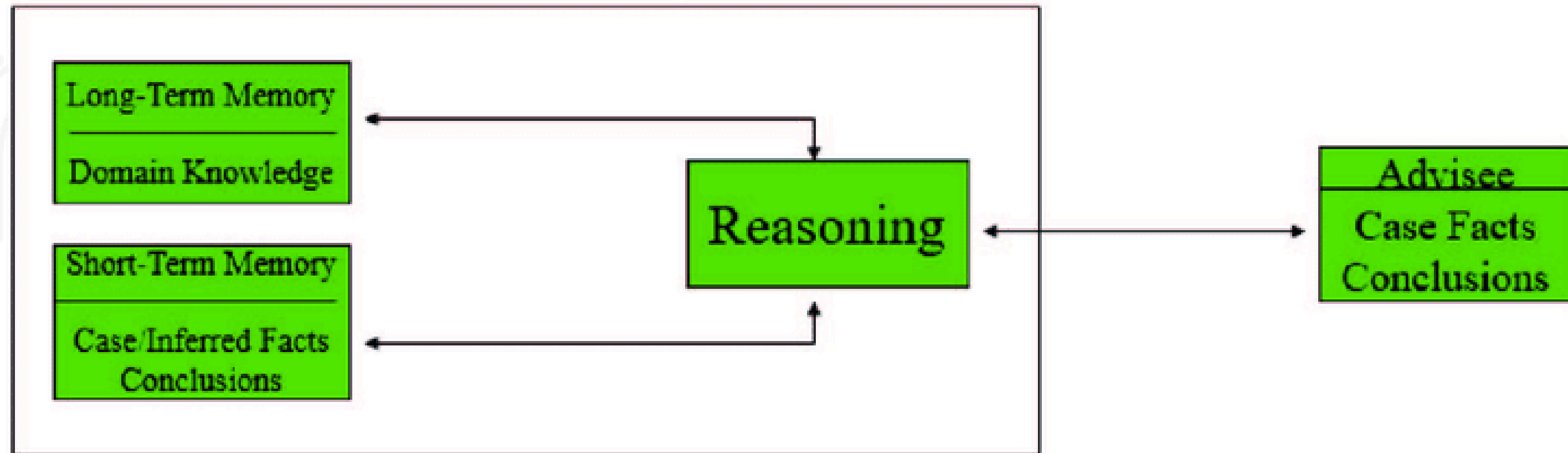
The Evolution of AI Through the Decades



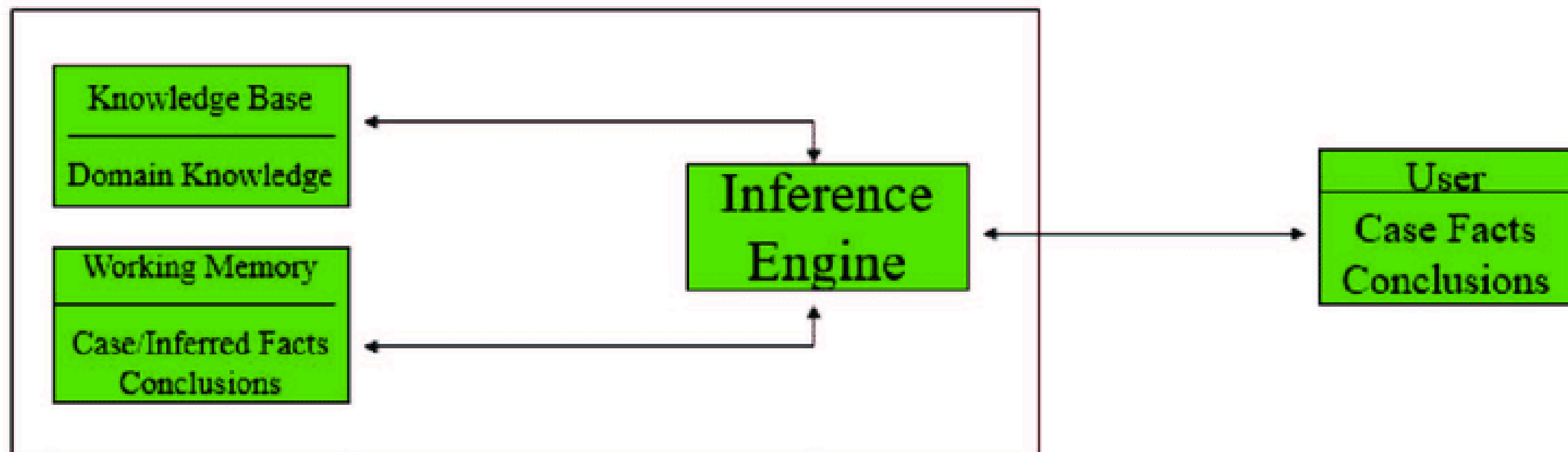
What are **expert systems**?

Computer programs that emulate the decision-making ability of human experts

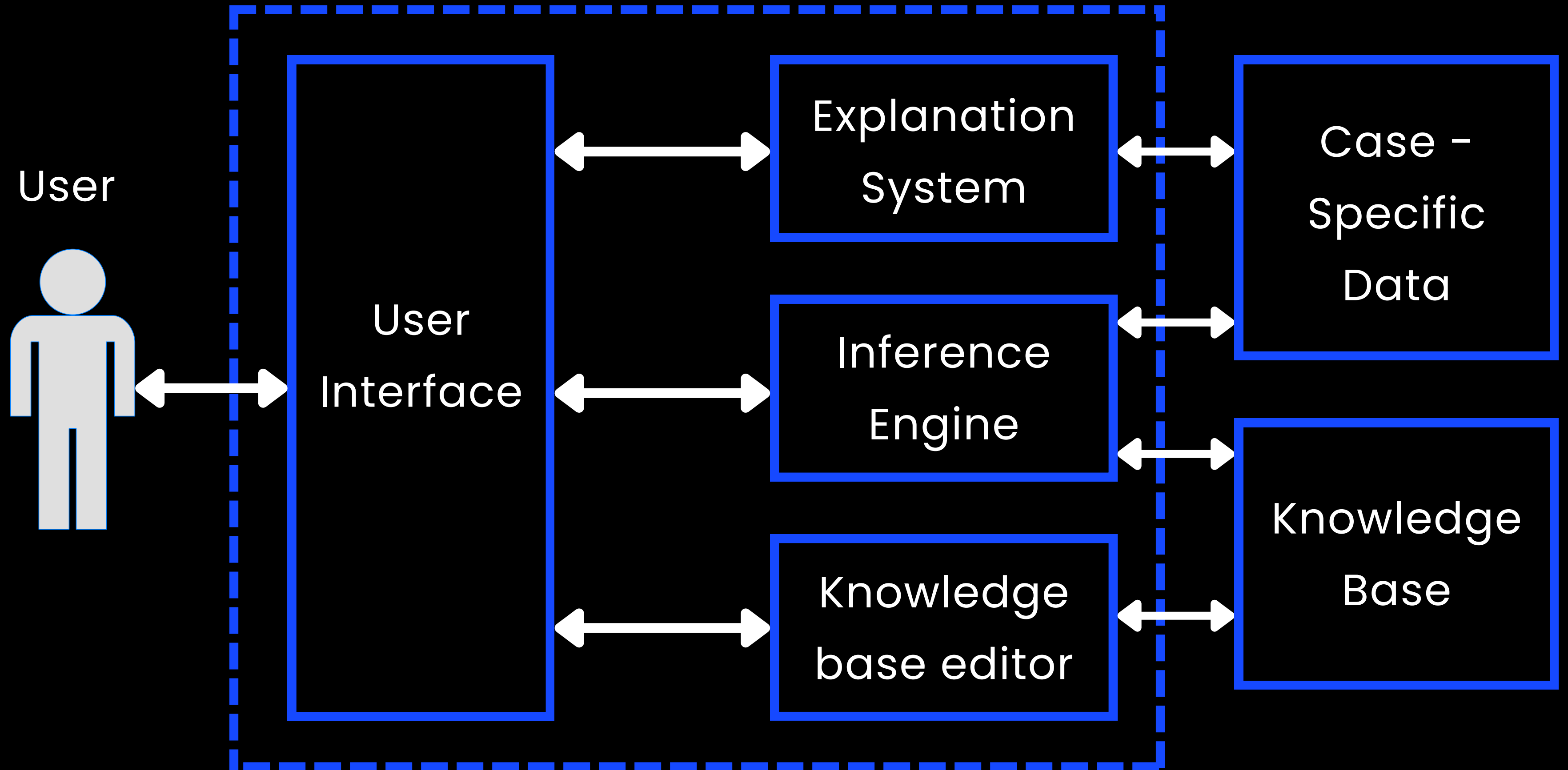
Human Expert



Expert System



The Architecture

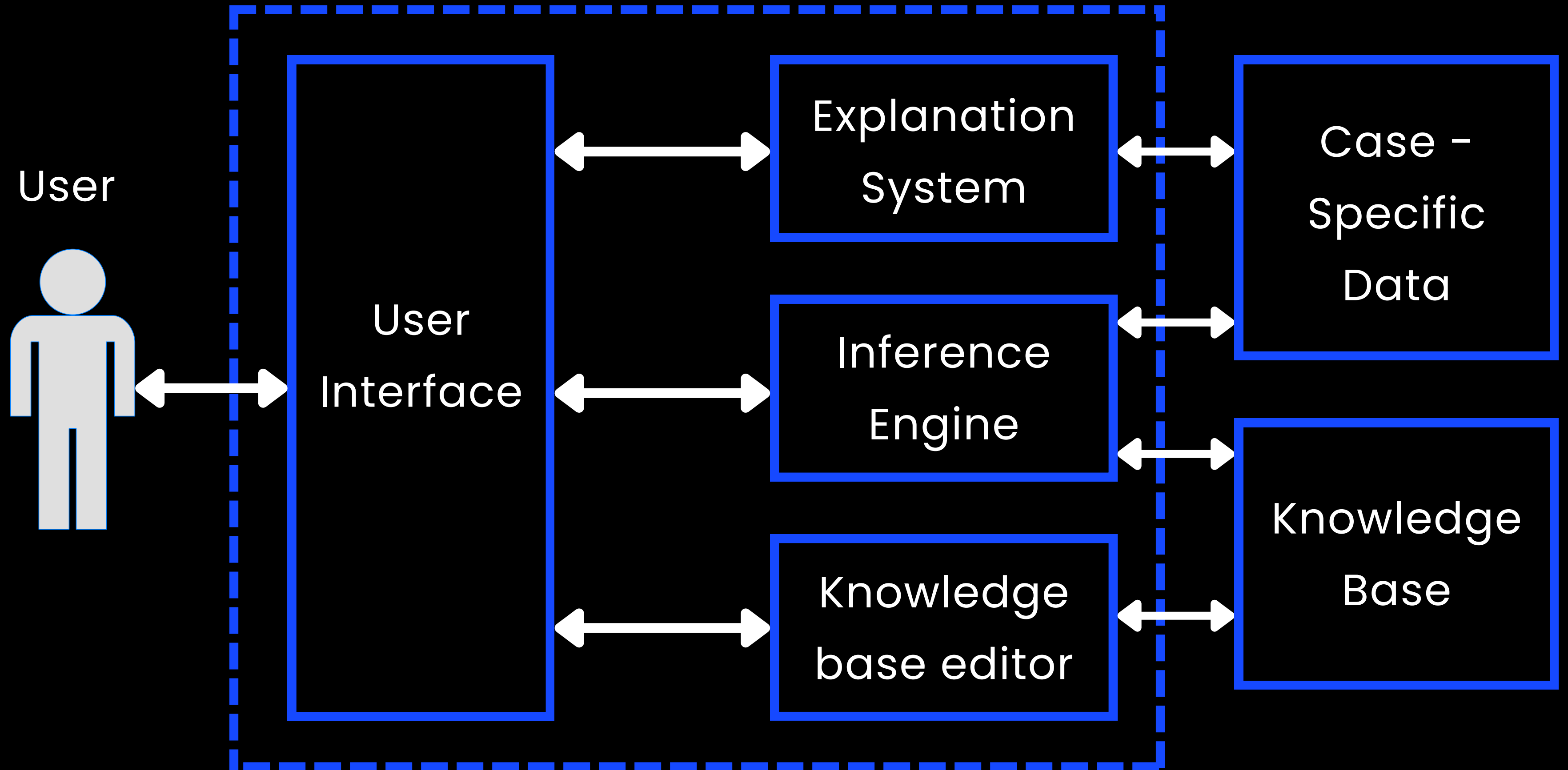


1. User Interface

- User interacts using menus / language
- Closely linked to explanation module
- User questions system and system questions user
- Two functions :- what question to ask next and keep previous questions record

Dialogue can be of 3 styles :-

- **System controlled** – where the system drives the dialogue through questioning the user
- **User controlled** – where the user drives the consultation by providing information to the system
- **Mixed control** – where both user and system can direct the consultation



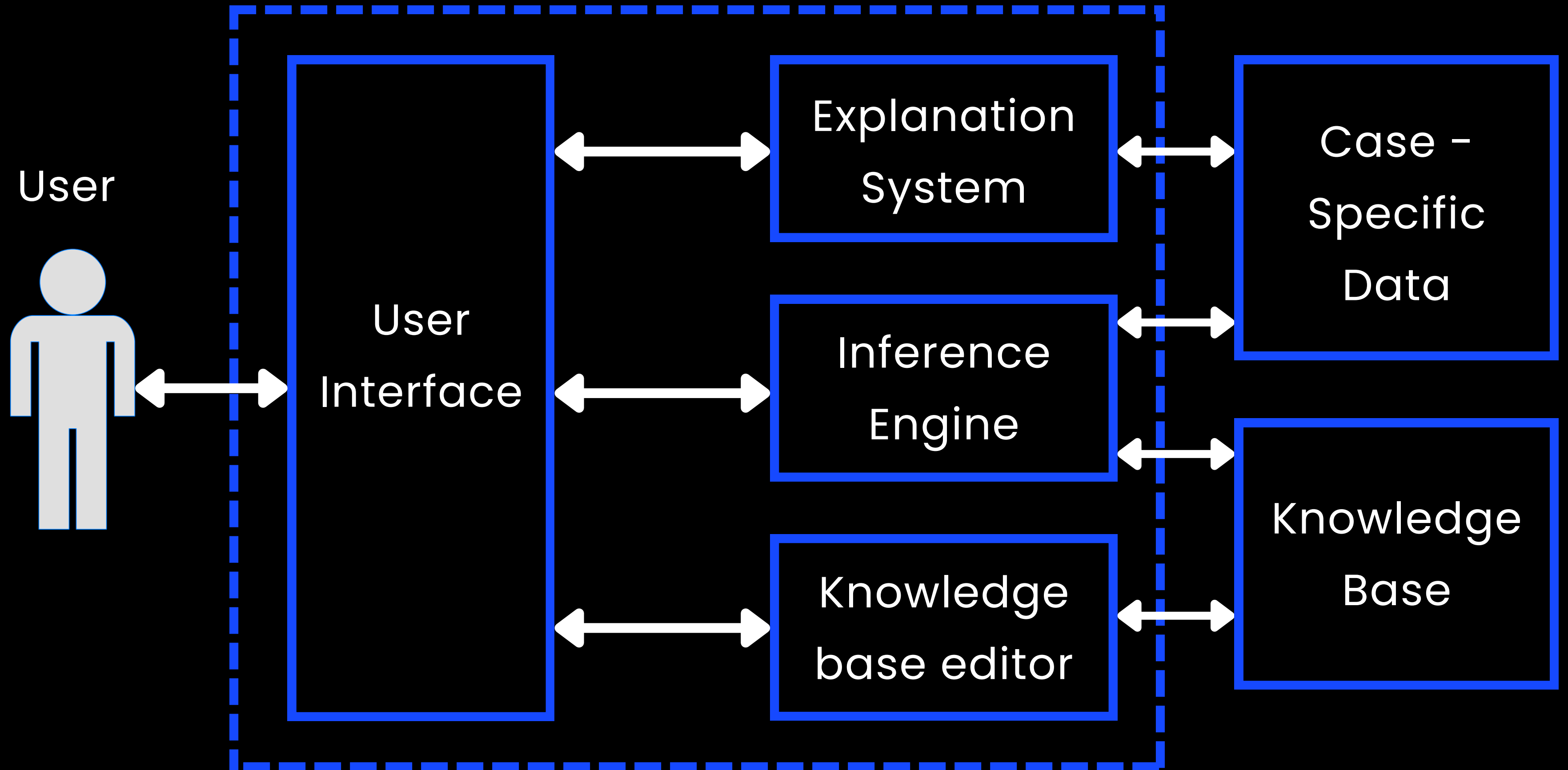
2. Knowledge base

- Contains both declarative knowledge and procedural knowledge
- Knowledge usually represented using rule-based production system approach
- If-then rules
- Meta rules

3. Inference Engine

- How and where to apply which knowledge
- Uses info provided by knowledge base and user to infer few facts and conclusion
- Determines which heuristic techniques
- Determines which rules are to be invoked

- Reasoning process can differ – forward chaining and backward chaining
- Knowledge and control are different.
Inference engine can be used with other knowledge bases.



4. Case specific data

- Data provided by user and partial conclusions based on the data

5. Explanation System

- Explains why it reached a certain conclusion or recommendation
- Users need to be convinced
- Makes possible to know that the system is working as desired

6. Knowledge base editor

- Mechanism for editing the knowledge base
- Syntax checking

